

Red Hat Network Management Command Line Guide

1. Basic Network Commands

- Show IP Address: `ip a`
- Show active interfaces: `ip link`
- Bring interface up: `ip link set eth0 up`
- Bring interface down: `ip link set eth0 down`
- Ping test: `ping 8.8.8.8`
- DNS test: `nslookup google.com`

2. Configure IP Address (Temporary)

```
ip addr add 192.168.1.100/24 dev eth0
ip route add default via 192.168.1.1
```

3. Configure IP (Permanent using nmcli)

- List connections: `nmcli con show`
- Edit a connection: `nmcli con edit eth0`
- Set static IP: `nmcli con mod eth0 ipv4.addresses 192.168.1.100/24`
- Set gateway: `nmcli con mod eth0 ipv4.gateway 192.168.1.1`
- Set DNS: `nmcli con mod eth0 ipv4.dns "8.8.8.8"`
- Set method manual: `nmcli con mod eth0 ipv4.method manual`
- Apply changes: `nmcli con up eth0`

4. Using nmtui (Text UI)

Command: `nmtui`

Use arrows to edit connections, hostname, etc.

5. Network Services Management

- Restart network: `systemctl restart NetworkManager`
- Enable on boot: `systemctl enable NetworkManager`
- Check status: `systemctl status NetworkManager`

6. Hostname Management

Command: `hostnamectl set-hostname myserver`

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7. View Routing Table

Command: `ip route`

8. Monitor Traffic

- Show open ports: `ss -tuln`
- Show listening services: `netstat -tulnp`
- Capture packets: `tcpdump -i eth0`
- Scan ports: `nmap <IP>`
- Show network path: `traceroute <host>`
- Live route monitoring: `mtr <host>`

9. DNS Configuration File

Edit file: `/etc/resolv.conf`

Add: `nameserver 8.8.8.8`

10. Firewall Management (firewalld)

- Check status: `firewall-cmd --state`
- Allow service: `firewall-cmd --add-service=http --permanent`
- Allow port: `firewall-cmd --add-port=8080/tcp --permanent`
- Reload firewall: `firewall-cmd --reload`